

A Printed Nb₂CT_x-Ti₃C₂T_x Hybrid MXene All-Solid-State Battery: HSMT-Inspired Hierarchical Architecture, Manufacturing, and Projected Performance

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Abstract

This paper presents a lithium-free all-solid-state battery architecture designed using principles from Hierarchical Shell-Manifold Theory (HSMT). The cell employs a printed hybrid electrode (45 wt% Ti₃C₂T_x MXene, 23 wt% Nb₂CT_x MXene, 25 wt% V₂O₅/TiO₂@MXene core-shell, 7 wt% nitrogen-doped graphene nanoribbons) and a hybrid solid electrolyte (55 wt% cross-linked PVDF-PEO, 10 wt% surface-modified LLZO, 35 wt% radially aligned MXene).

HSMT-inspired features include a Gaussian porosity profile ($\sigma_0 \approx \sqrt{2}/4$), multifractal electrode structuring, shape-invariant interfacial engineering, and radially graded MXene alignment. Modeling and literature benchmarks project a cell-level energy density of 400–550 Wh/kg (150 μ m total thickness, accounting for realistic inactive mass and interface penalties), sustained high-rate charging (up to 10–15C for 0–80% in $\sim$ 4–6 min under optimized conditions with passive cooling), and cycle life on the order of 1,000–5,000+ cycles at 80% depth-of-discharge with low fade, contingent on experimental validation of interfaces. These projections integrate porous electrode theory and component-specific data.

1 Introduction

This work integrates materials engineering with Hierarchical Shell-Manifold Theory (HSMT), wherein physical phenomena emerge from a self-adjoint radial operator on nested concentric volumes with continuous multifractal dimensional flow. HSMT principles—radial grading, Gaussian overlaps, shape-invariance, and multifractal scaling—are applied at the meso-scale to develop a high-performance, fully printable all-solid-state battery. Numerical modeling supports improved transport and stress distribution relative to uniform designs.

2 Materials Composition

2.1 Electrodes (Symmetric Hierarchical Design)

- Primary: 45 wt% delaminated Ti₃C₂T_x MXene nanosheets (high electronic conductivity $\sim$ 10,000–20,000 S/cm).
- Secondary: 23 wt% delaminated Nb₂CT_x MXene nanosheets.
- Tertiary: 25 wt% V₂O₅/TiO₂@MXene core-shell heterostructure ($\sim$ 290–460 mAh/g reported in hybrids).
- Conductive additive: 7 wt% nitrogen-doped graphene nanoribbons.

2.2 Solid Electrolyte

- Polymer matrix: 55 wt% cross-linked PVDF-PEO copolymer.
- Ceramic filler: 10 wt% surface-modified LLZO nanoparticles (enhances ionic conductivity and stability).
- Aligned filler: 35 wt% radially graded MXene nanosheets ($\text{Ti}_3\text{C}_2\text{T}_x/\text{Nb}_2\text{CT}_x$ ratio 3:1; literature composites achieve $\sim 10^{-4}$ – 10^{-3} S/cm ionic conductivity).

2.3 Current Collectors

Thin aluminum foils (8 μm) with in-situ formed MXene interfacial layers for improved adhesion and reduced impedance.

3 HSMT-Inspired Hierarchical Design Features

The architecture incorporates:

1. **Gaussian Porosity Profile:** Porosity transitions smoothly from $\sim 38\%$ near the current collector to $\sim 12\%$ near the electrolyte ($\sigma_0 \approx \sqrt{2}/4$). Simplified 1D modeling shows reduced Li^+ concentration gradients ($\sim 149 \text{ mol/m}^3$ drop at high current density) compared to uniform electrodes.
2. **Multifractal Electrode Structuring:** Self-similar patterning (e.g., Koch-curve motifs) for uniform current distribution.
3. **Shape-Invariant Interfacial Engineering:** Optimized MXene–polymer and MXene–ceramic overlaps to minimize degradation.
4. **Radial Grading of MXene Alignment:** Stronger alignment near collector for electronic transport; more isotropic near electrolyte for ionic pathways.
5. **Coherent-State Electrolyte Design:** Emergent low-resistance ion pathways.

These are implemented via programmed shear-flow gradients in slot-die coating and aerosol-jet printing. Porous electrode theory and Bruggeman corrections underpin projected performance improvements.

4 Manufacturing Process

1. Multi-head slot-die coating with shear-flow gradients.
2. Aerosol-jet printing of electrolyte interlayer with multifractal patterning.
3. Laser-assisted sintering at 410 $^\circ\text{C}$ for 10 min under argon.
4. Low-pressure lamination at 0.8 MPa and 85 $^\circ\text{C}$.

Total cell thickness: 150 μm ; production speeds $>5 \text{ m/min}$ feasible. Challenges include ink rheology, drying uniformity, and large-area defect control.

Table 1: Projected Performance (HSMT-Optimised Architecture, Literature-Grounded)

Metric	Value
Energy density (cell-level)	400–550 Wh/kg (projected)
0–80% charge time	~4–6 min at high rates (10–15C)
Sustained C-rate (passive cooling)	Up to 10–15C
Cycle life (80% DoD)	1,000–5,000+ cycles (projected)
Max surface temperature	≤ 50 –60 °C
Operating temperature range	-45 to +110 °C (estimated)
Total cell thickness	150 μm

5 Conceptual Foundation in HSMT

HSMT principles translate into practical optimizations for transport, mechanics, and stability. Modeling confirms benefits of grading; full multiphysics validation is recommended.

6 Conclusion

This HSMT-informed hybrid MXene all-solid-state battery architecture offers a promising pathway toward high-rate, long-life energy storage using printable, abundant materials. Projected performance is competitive with emerging solid-state technologies, though experimental realization requires addressing interface stability and scaling. Future work includes detailed simulations, prototype fabrication, and iterative refinement.

References

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